

Arijit Chanda

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Education

Indian Institute of Technology (Indian School of Mines) Dhanbad

Bachelor of Technology in Electrical Engineering

Expected May 2026

Dhanbad, Jharkhand

- **Relevant Coursework:** Data Structures and Algorithms (C++), Operating Systems, Object Oriented Programming, Database Management Systems.

Projects

IntelliDoc: Your Health Buddy | *React.js, MongoDB, Node.js, Express.js, Flask, OpenCV, Tensorflow*

[GitHub](#)

- **Developed** a web application enabling users to input health data via images or manual entry, generating personalized diagnostic reports and analytical charts with **90%+ accuracy**, based on user tests.
- **Strengthened** user experience by creating MediBuddy, an interactive chatbot that provided tailored health advice to **50+ users** during testing.
- **Integrated** a machine learning engine to analyze health data, enabling proactive identification of risks across **10+ conditions** and strengthening early detection for test users.

Movie Recommender System | *Flask, TensorFlow, scikit-learn*

[GitHub](#)

- **Architected** a content-based movie recommender system using the **TMDB 5000** dataset, achieving an **85%+ accuracy** rate for personalized suggestions.
- **Designed** a composite feature representation for each movie by extracting and weighting key metadata, including genre, director, plot keywords, and the **top 4** principal cast members.
- **Implemented** a **Bag-of-Words (BoW)** model using Scikit-learn's CountVectorizer to transform the processed text into a high-dimensional vector space, resulting in a **4800x5200** sparse matrix.
- **Optimized** the system to generate real-time recommendations in less than **1 second**.

Martian Landmark Detection using CNNs | *OpenCV, PyTorch, TensorFlow, scikit-learn*

[GitHub](#)

- **Engineered** a deep learning model using TensorFlow and transfer learning, boosting model generalization with data augmentation to achieve a train and validation **accuracy of over 92%** on Martian landmark classification.
- **Improved** model performance, by implementing **data augmentation** strategies that addressed previous challenges with inconsistent lighting in the orbital imagery dataset.
- **Automated** pipeline to process orbital imagery with OpenCV, classify **5+ distinct** Martian landmarks, and visualize the findings with Matplotlib.

Technical Skills

Languages: C, C++, Python.

Technologies: TensorFlow, PyTorch, Keras, Flask, scikit-learn, Matplotlib, Numpy, Pandas, Git/GitHub, OpenCV.

Concepts: Artificial Intelligence, Machine Learning, Neural Networks, Deep Learning, Natural Language Processing (NLP), LLM, OOPs (Object-Oriented Programming Systems), Data Structures, and Algorithms (DSA).

Core Competencies: Leadership, Event Management, Communication.

Programming Achievements

- Achieved a **1322** (Pupil) peak rating on Codeforces in competitive programming contest.
- Attained a maximum Codechef rating of **1742** (3 stars) from consistent contest performance.
- Secured **Global Rank 202** in Global starters 124 on Codechef among 10000+ participants.
- Solved **500+ questions** across various coding platforms and online judges.
- Honed problem-solving and algorithmic skills by solving **250+ LeetCode questions**.

Social Engagements

Head of Promotions: Parakram'24 (Sports-fest) and Concetto'24 (Techno-Management Fest).

Club Member: FotoFreaks – Photography Club, IIT (ISM) Dhanbad.

Sports Engagements: Football, Chess.

Volunteer: Kartavya (Student-run NGO) – Educating underprivileged children.